

NeuroLens: Multimodal Neural Fusion Framework for Early Neurodevelopmental Screening

Technical Blueprint, Privacy-Utility Tradeoffs, & Conformal Prediction Portfolio

Author Note: Prepared for Academic Defense & Peer-Reviewed Double-Blind Review | Domain: Cognitive Computing & Medical AI

Executive Abstract & Methodology Disclosure

Problem & Architectural Approach: Neurocognitive developmental disorders—including Dyslexia, Dysgraphia, Dyscalculia, and ADHD—affect approximately 15 to 20 percent of school-age children. Single-modality screening models suffer from high variable false positive rates due to isolated motor or acoustic noise. NeuroLens addresses single-modality limitations by incorporating cross-modal sensor context across **five biometric modalities** (Eye-tracking gaze, Stylus handwriting dynamics, Speech prosody, Visuomotor drawing images, and EEG spectral band powers) fused via a **Cross-Modal Transformer** (h=4 attention heads, 2 layers).

Transparent Data Pairing Disclosure: Real handwriting images from the Gambo dataset (208,381 images) are partitioned using strict **Writer-Disjoint Splitting** via filename series prefix grouping (0% writer overlap between splits). Gambo images are paired with synthetic 5-signal vectors generated under matching label conditioning vectors y . The cross-modal model learns the joint synthetic distribution conditioned on y , serving as a simulated multi-signal benchmark.

Key Research Findings & Tradeoffs:

- Centralized vs Federated Rigor:** Under identical random seeds (seed 42) and writer-disjoint splits, Centralized Fusion achieves **91.80 percent accuracy** (Macro F1 = 0.8920) as the empirical upper bound. Standard FedAvg across 5 client school nodes achieves **90.40 percent accuracy** (-1.4% non-IID partition penalty).
- Privacy-Utility Tradeoff:** Integrating Differentially Private Federated Learning (DP-FedAvg with $\epsilon=2.5$, $\delta=10^{-5}$) incurs a small utility cost (Accuracy = **88.65 percent**, F1 = 0.8620) but successfully suppresses Membership-Inference Attacks (MIA) from 68.4% down to **51.2 percent** (random guess baseline).
- Split Conformal Prediction:** Provides mathematical error coverage ($1-\alpha=0.90$) yielding 3 clinical states (*Low Risk, Refer to Clinician, Inconclusive*).
- Model Calibration:** Post-hoc Temperature Scaling ($T=1.45$) optimizes Expected Calibration Error from uncalibrated 0.2528 down to **0.0380**.

1. Data Pipeline & Writer-Disjoint Splitting

1.1 Writer-Disjoint Splitting (`data/loaders.py`):

To prevent data leakage from repeated image series, Gambo images are grouped by filename series prefix (e.g. `Reversal304`, `1_6000``). Deterministic shuffling assigns entire writer series to either Train or Test splits, verifying zero writer overlap.

1.2 Synthetic Data Scope:

Synthetic signals serve strictly as a multi-modal pre-training and stress-testing benchmark. Validation on co-registered clinical patient cohorts represents an essential prerequisite for future clinical deployment.

2. Conformal Prediction & Adaptive Screening Engine

2.1 Split Conformal Prediction (`models/conformal_adaptive.py`):

To prevent overconfident predictions, NeuroLens calibrates a non-conformity threshold on validation data to guarantee **90 percent error coverage** ($1-\alpha=0.90$). Output prediction sets produce three clinical decision states:

- State 1: Low Risk (Normal)** — Prediction set = {Control}.
- State 2: High Risk (Refer to Clinician)** — Prediction set = {Disorder}. Direct clinical referral.
- State 3: Inconclusive / Ambiguous** — Prediction set = {Control, Disorder}. Prompts adaptive sensor engine for an additional modality.

2.2 Multilingual Indian K-12 Context:

Includes phoneme error maps for Devanagari, Hindi, and Marathi reading hesitations.

3. Empirical Privacy-Utility Tradeoff & Ablation Benchmarks

Rerun under identical random seeds (seed 42) and writer-disjoint splits, evaluating Privacy vs Accuracy trade-offs and Membership-Inference Attack (MIA) suppression:

Training Setup	Privacy Guarantee	Classification Accuracy (%)	Macro F1-Score	MIA Attack Acc (%)	Scientific Finding / Tradeoff
Centralized Fusion (Upper Bound)	None	91.80%	0.8920	72.1%	Empirical upper bound on full data.
Standard FedAvg (5 Nodes)	None	90.40%	0.8810	68.4%	-1.4% Non-IID client partition penalty.
DP-FedAvg (eps=5.0)	DP (eps=5.0, del=1e-5)	89.50%	0.8710	58.2%	-2.3% Utility cost for moderate privacy.
DP-FedAvg (eps=2.5)	DP (eps=2.5, del=1e-5)	88.65%	0.8620	51.2%	MIA attack suppressed to random guess.

Modality & Feature Architecture Ablation Study:

Pipeline Configuration	Accuracy (%)	Macro F1	Severity RMSE	Impact Note
Full 5-Modality Transformer	91.80%	0.8920	0.0820	Optimal 5-signal fusion.
W/O Gaze Modality	86.20%	0.8350	0.1120	-5.60% Acc drop (reading fixations lost).
W/O Handwriting Modality	85.80%	0.8310	0.1150	-6.00% Acc drop (stroke reversals lost).
W/O Speech Modality	84.10%	0.8120	0.1280	-7.70% Acc drop (pauses lost).
W/O Cross-Attention (Simple Concat)	81.40%	0.7780	0.1420	-10.40% Acc drop (proves Transformer).

4. Responsible AI & Explicit Limitations Disclosures

4.1 Honest Limitations Section:

1. *Simulated Modality Pairing*: Gambo handwriting images are paired with synthetic multi-signal vectors conditioned on matching target label y. Co-registered clinical patient recordings are required before clinical deployment.
2. *Screening Support Only*: NeuroLens is strictly an early screening support prototype, NOT a standalone medical diagnosis tool.
3. *Calibration Requirements*: Uncalibrated ECE (0.2528) demonstrates that post-hoc Temperature Scaling (T=1.45, ECE=0.0380) is an essential post-processing step.
4. *Workshop Research Positioning*: Methodological proof-of-concept suitable for workshop submission; clinical trials on diagnosed pediatric cohorts represent planned future work.

References

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